

WORK ORDER EXECUTION & OVERHAUL REPORT — USER-TEST-001

Work Order Number: WO-8812

Equipment Tag: USER-TEST-001

Equipment Name: User Test Pump (Centrifugal Process Pump)

Facility / Area: Gulf Coast Refining — Unit 2 Fluid Transfer

Maintenance Date: 2026-08-24

Technician: Marcus Vance (Lead Mechanical Reliability Specialist)

Executive Summary:

Comprehensive overhaul and corrective maintenance performed on centrifugal pump USER-TEST-001 following scheduled preventative maintenance trigger.

Detailed Scope of Work:

1. Isolated pump USER-TEST-001 mechanically and electrically via lockout/tagout protocol LOTO-4491.
2. Disassembled drive-end and non-drive-end bearing housings.
3. Removed degraded bearings and replaced with genuine SKF-6314-2Z deep groove ball bearing (DE) and SKF-NU-314 cylindrical roller bearing (NDE).
4. Inspected mechanical seal cartridge BURG-M7N; replaced elastomeric secondary O-rings and flushed seal faces.
5. Reassembled pump casing and performed precision laser shaft alignment. Angular misalignment was adjusted from 0.12 mm down to 0.02 mm (well within OEM tolerance of 0.05 mm).
6. Charged bearing cavity with 180 grams of Mobil SHC Polyrex 462 synthetic grease.

Final Test Run & Handover:

Restarted pump USER-TEST-001 under full process load (145 m³/h at 2950 RPM). Overall vibration RMS measured 1.45 mm/s at drive end, significantly below ISO 10816-3 alert limit of 4.5 mm/s. Bearing temperatures stabilized at 62.4 C.

Pump USER-TEST-001 certified for continuous unconstrained service.